

Dear Foundation EGI Hiring Team,

I am excited to apply for the Research Scientist position. With a Ph.D. in Computational Mechanics and strong experience in code development for geometry processing, simulation, and data-driven modeling, I am eager to contribute to your AI Copilot for design and manufacturing.

At Los Alamos National Laboratory, I have designed, developed, and maintained production-grade surrogate modeling pipelines and Python tooling that transform thousands of high-fidelity physics simulations (CAD/CAE data) into fast, engineering-scale predictions. This work involved geometry-aware data processing, dataset curation and ablation studies, physics-informed algorithms, optimization techniques, and integration of Python workflows with large C++ simulation frameworks (MOOSE/BISON). I also have direct experience with inverse methods (based on digital image correlation) for full-field geometry and material parameter identification from experimental data.

I am particularly drawn to Foundation EGI's mission of combining advanced geometry processing, simulation, and AI to transform real engineering workflows. Your focus on turning complex CAD/CAE/CAM data into structured inputs for learning systems, along with post-training and domain-specific language development, aligns closely with my background in bridging detailed physics simulation with practical, deployable engineering tools.

I would welcome the opportunity to discuss how my experience in simulation algorithms, geometry processing, data pipelines, and code development can support Foundation EGI's work.

Thank you for your consideration.

Warm regards,

Andre Ruybalid

andrerybalid@gmail.com | +1 719 210 1746